National University of Computer and Emerging Sciences

FAST School of Computing **Spring-2023** **Islamabad Campus**

EE-1005: Digital Logic Design

Serial No:

Sessional Exam-I

Total Time: 1 Hour

Total Marks: 50

Tuesday, 28th February, 2023

Signature of Invigilator

Course Instructors

Dr. Mehwish Hassan, Dr. Niaz Ahmed, Shehzad Ahmed, Dr. Muhammad Awais, Nirmal Tariq

Areeq Ahmed Malik — Student Name
22i-1167 — Roll No.
CS-B — Course Section
Student Signature

DO NOT OPEN THE QUESTION BOOK OR START UNTIL INSTRUCTED.

Instructions:

1. Attempt on question paper. Attempt all of them. Read the question carefully, understand the question, and then attempt it.
2. No additional sheet will be provided for rough work. Use the back of the last page for rough work.
3. If you need more space write on the back side of the paper and clearly mark question and part number etc.
4. After asked to commence the exam, please verify that you have **Nine (9)** different printed pages including this title page. There are a total of **Four (4)** questions.
5. Calculators are not allowed for computations.
6. Use permanent ink pens only. Any part done using soft pencil will not be marked and cannot be claimed for rechecking.

	Q-1	Q-2	Q-3	Q-4	Total
Marks Obtained	15	9	12	8.5	44.5
Total Marks	15	13	12	10	50

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Question 1 [15 Marks]

1. Convert $(A0A)_{16}$ to octal representation. [2]

Solution:

$(A0A)_{16} \rightarrow (1010\ 0000\ 1010)_2$

$(A0A)_{16} \rightarrow \underline{(5012)_8}$

2. Convert $(A0A)_{16}$ to decimal representation. [2]

Solution:

By using sum of weight

$(16^2 \times 10) + (16^1 \times 0) + (16^0 \times 10)$

$2560 + 0 + 10$

$(2570)_{10}$

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3. In a 8-bit two's-complement system, what decimal number does the bit pattern 10000111 represent? [3]

Solution:

Number:- 10000111

2's Comp:- 01111001

↓

(121)

So, 10000111 → $(-121)_{10}$

4. One of the following bit patterns is valid BCD (binary-coded decimal), but the other one is not: 100110110100, 100100111000. Which one is not valid? For credit to be given, you must give a correct reason. [1]

Solution:

First Number: 1001 1011 0100

↓

11 which is an invalid BCD code.

So, first number is not valid BCD.

5. What number does the valid bit pattern from part (4) represent? Give your answer in base ten. [1]

Solution:

Valid BCD :- 1001 0011 1000

$(9\ 3\ 8)_{10}$

6. The ten-bit Gray code for 353_{10} is 0111010001. Explain briefly but precisely why it cannot possibly be true that 0111010100 is the ten-bit Gray code for 354_{10}. Or calculate gray code for 354_{10}. [1]

Solution:

~~354 = (0111010010)2~~

353 = 0111010001

354 = 0111010100

This gray code is invalid because consecutive gray codes should only have a one bit change, and This code has 2 bit change.

7. Add BCD numbers 256_{10} and 464_{10}. [2]

Solution:

256 = 0010 0101 0110

464 = 0100 0110 0100 +

0111 1100 1010

+0110 +0110 6

0111 0010 0000

7 2 0

256+464 = (720)

8. Find Subtraction of $(402)_8$ and $(314)_8$ using 7's complement method. [3]

Solution:

$(402)_8 - (314)_8$

7's Comp of 314 =

777
314
463

402
463
1065
+1
$(66)_8$

402
464
1066

8|13
1-5

8|8
1-0

~~408~~
~~+ 465~~
~~1075~~
~~+1~~
~~discard carry~~
~~$(075)_8$~~
~~$(676)_8$~~

Question 2 [13 Marks]

1. **Convert the given function to form the Sum of Minterms.** **[3]**

$$F(A,B,C,D) = \prod(0,3,4,5,9,11,14)$$

Solution:

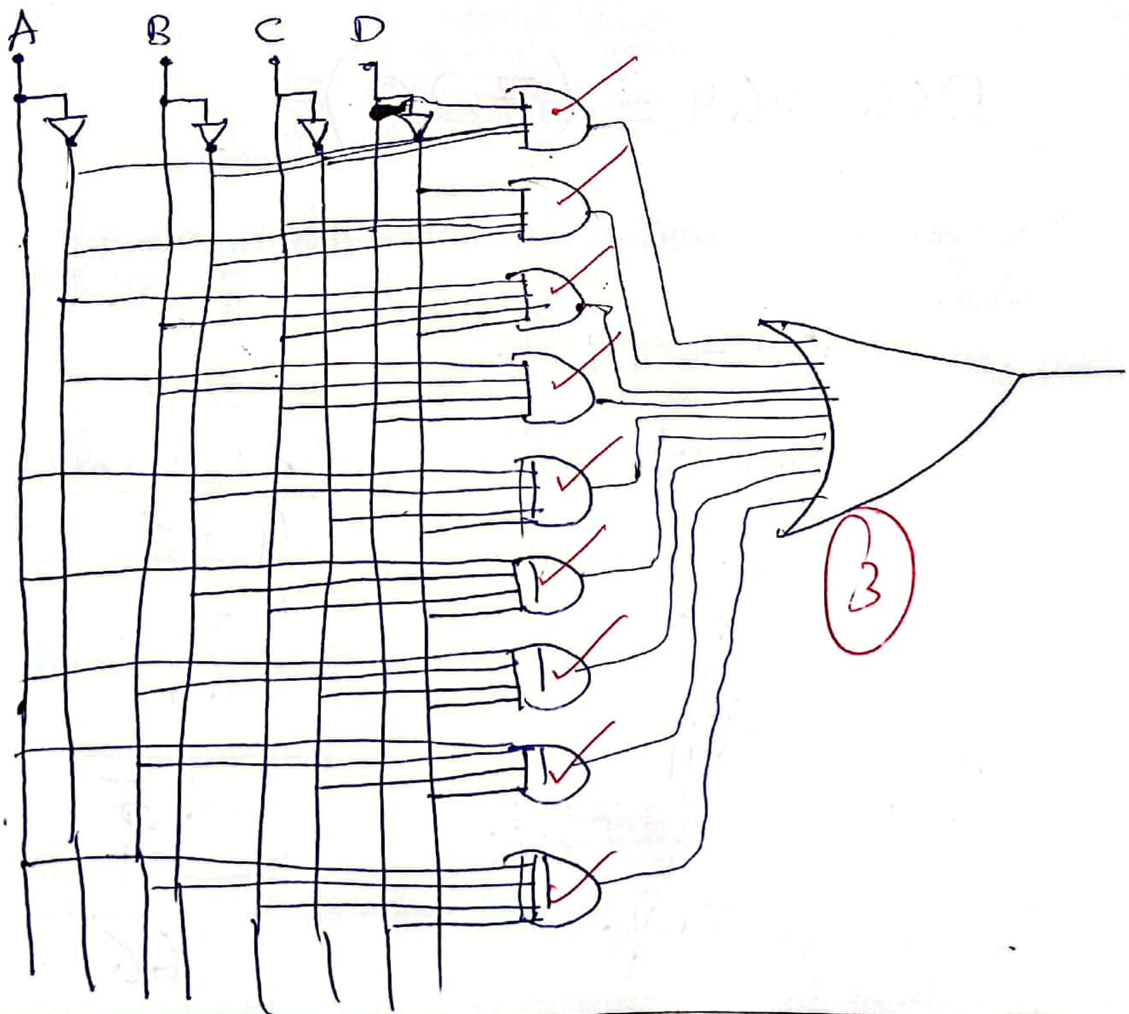
$$F(A,B,C,D) = \sum(1,2,6,7,8,10,12,13,15)$$

$$F(A,B,C,D) = \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}BC\bar{D} + \bar{A}BCD + A\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D} + AB\bar{C}\bar{D} + AB\bar{C}D + ABCD$$

3

2. **Draw the circuit diagram from the Canonical SOP form written in part (1).** **[3]**

Solution:



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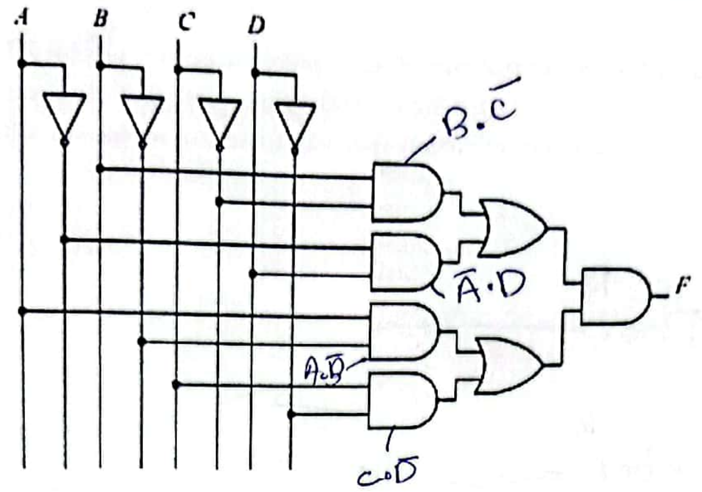
3. Write the expression for the given circuit and reduce using Boolean Algebra. **[5]**



Solution:

$$F = \left((B\cdot\bar{C}) + (\bar{A}\cdot D)\right) . \left((A\cdot\bar{B}) + (C\cdot\bar{D})\right)$$

$$F = (B\cdot\bar{C})(A\cdot\bar{B}) + (B\cdot\bar{C})(C\cdot\bar{D}) + (\bar{A}\cdot D)(A\cdot\bar{B}) + (\bar{A}\cdot D)(C\cdot\bar{D})$$

3

$$F = \bar{A}\cdot D(C\cdot\bar{D} + A\cdot\bar{B}) + B\cdot\bar{C}(C\cdot\bar{D} + A\cdot\bar{B})$$

$$\cancel{F = (C\cdot\bar{D} + A\cdot\bar{B})}$$

4. Draw the reduced circuit diagram found in part (3). **[2]**

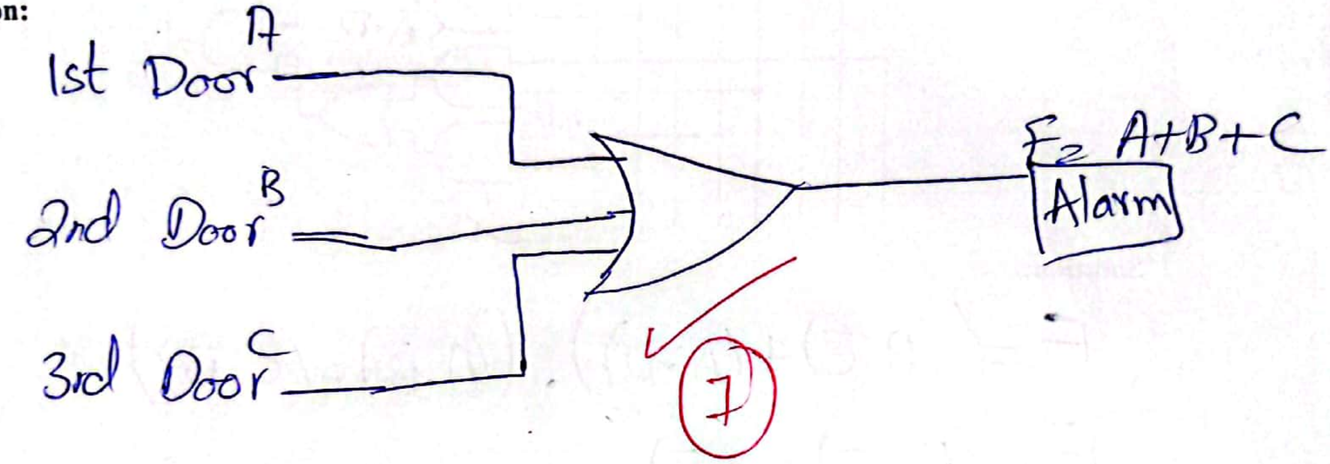
Solution:

× 0

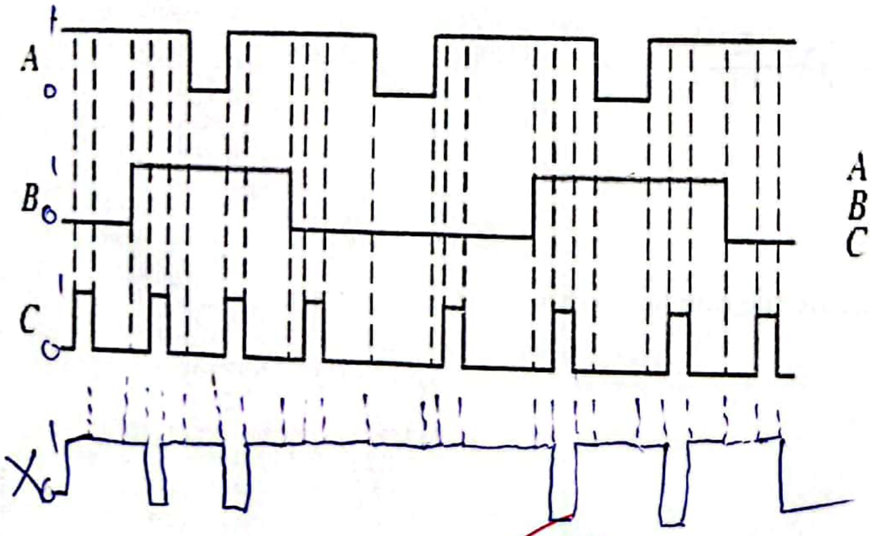
Question 3 [12 Marks]

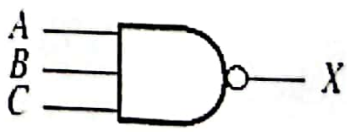
1. A residential house has three entrances. It is required to secure all the entrances with magnetic sensors. Magnetic sesnors output a logic LOW when door is closed and a logic HIGH when any door opens. Design a simple circuit that can generate an alarm when any of the door is intruded. [7]

Solution:



2. Draw the output of the given circuit. [5]





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Question 4 [10 Marks]

1. Use a Karnaugh map to minimize the following function into minimal SOP expression:

$(\bar{A}+\bar{B}+C+D)(A+\bar{B}+C+D)(A+B+C+\bar{D})(A+B+\bar{C}+\bar{D})(\bar{A}+B+C+\bar{D})\ (A+B+\bar{C}+D)$

Solution:

$F = \Pi(12, 4, 1, 3, 9, 2)$

$F = \Sigma(5, 6, 7, 8, 10, 11, 13, 14, 15)$.

AB \ CD	00	01	11	10
00	1			
01		1	1	1
11		1	1	1
10	1		1	1

$F = A\overline{B}\overline{D} + AC + BC + BD$

8.5